

VELIDANDA KRISHNA SAI

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Education

International Institute of Information Technology, Bangalore

Jul 2023 - Jul 2028

Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering

CGPA: 3.51/4.00

Experience

Undergraduate Research Affiliate - Multimodal Stress Detection

Jan 2026 - May 2026

Samsung Lab, IIITB | Supervisor: Dr. Sakshi Arora

Bangalore, India

- Designed a multi-stressor protocol, recording 8-channel EEG (500 Hz) and ECG from 15 subjects.
- Engineered an end-to-end **data processing pipeline** to extract, clean, and transform raw 8-channel EEG (500 Hz) and ECG signals into a 4,939-window feature set.
- Trained **SVM** and **CNN-LSTM** classifiers with Leave-One-Subject-Out CV, achieving **95.6%** subject-independent stress detection accuracy.

Achievements

- Finalist:** NCIIPC AI Grand Challenge - Emerged as one of the top 3 teams (Coders@IIITB) - 100,000+ teams. [🔗](#)
- First Prize:** ICDEC'24 International Vehicle Detection Challenge - 1st of 2,000 teams. [🔗](#)
- National Runner-Up:** TruthTell Hackathon (WAVES Summit 2025) - Top 2 of 5,600+ global submissions. [🔗](#)
- 3rd Runner-Up:** Google Gen AI Exchange Hackathon 2025 - 4th among 270,000+ developers. [🔗](#)
- Amazon ML Summer School:** Selected among the top 3,000+ students chosen from over 100,000 applications nationwide.
- Dean's Merit List:** Named to the DML for students with GPA of 9 or above for Academic Year 2023-2024. [🔗](#)
- JEE Mains:** Secured a rank of **4,439** out of over 1,100,000 applicants.

Projects

VoltIQ: EV Fleet Intelligence [🔗](#) | Python, FastAPI, MQTT, Agentic AI

Jun 2026 - Jul 2026

- Designed and deployed a scalable **real-time data pipeline** using an MQTT broker and **FastAPI** to ingest continuous fleet telemetry data into **TimescaleDB**.
- Engineered automated data transformation workflows that feed live telemetry into **PyTorch** models, enabling real-time recomputation of RUL/SoH metrics.
- Implemented graph route optimization (A^* /ALT) and a **multi-agent CrewAI** service for automated work orders.

HFT Simulator Engine [🔗](#) | C++, Python, Data Structures and Algorithms

Jun 2026 - Jul 2026

- Built a high-throughput **market data ingestion pipeline** utilizing a **lock-free SPSC** architecture for processing large volumes of simulated order data.
- Implemented custom data structures, including a Robin Hood hash map and pool allocator, to achieve **O(1)** latency in order processing and cancellation.
- Developed a **C++** order matching engine with intrusive FIFO queues, optimizing for rapid data state changes and partial fills.

Relay Brain [🔗](#) | FastAPI, Python, TypeScript, AI Agents, MCP, Open-Source

May 2026 - Jun 2026

- Created **Relay Brain**, a context-intelligence layer that orchestrates multiple AI agents across a project via an **MCP** server, maintaining shared state and task history across agent handoffs.
- Formulated an **LLM-based** reasoning core that decomposes goals into structured task plans and routes subtasks to the right agent based on live project memory.
- Delivered a **FastAPI** backend and **Next.js** mission-control dashboard for real-time debugging and visibility into agent decisions.

Dialecto [🔗](#) | FastAPI, MongoDB, Python, REST APIs, GenAI, Agentic Workflows

Nov 2024 - Dec 2024

- Architected an asynchronous full-stack backend using **FastAPI** and **RESTful APIs**, implementing modular MVC service layers, **Pydantic** validation schemas, and robust session handling for seamless client-server state synchronization.
- Engineered flexible document data models in **MongoDB**, leveraging **compound indexing** and **aggregation pipelines** to optimize read/write throughput for multi-turn chat history, dynamic game states, user streaks, and live leaderboards.
- Engineered an **agentic AI engine** delivering real-time grammar feedback and adaptive difficulty progression.

Technical Skills

Languages: Java, C++, C, Python, JavaScript, SQL

Databases & Data Tech: PostgreSQL, MySQL, MongoDB

Technologies & Frameworks: FastAPI, Docker, Kubernetes, Git, PyTorch, LangChain, CrewAI

DevOps: Docker, CI/CD (Jenkins, GitHub Actions), Kubernetes

Cloud: Google Cloud Platform, Amazon Web Services

Certifications: Google Cloud Platform Certification for Data Science, AI and ML. [🔗](#)

Relevant Coursework

Data Structures & Algorithms, Object Oriented Programming (C++ & Java), Operating Systems, Computer Networks, Database Systems, Recommendation Systems, Natural Language Processing, Software Engineering, Machine Learning

Extracurricular

- ICP Crewsphere IIITB Chapter (Core Member):** Orchestrated technical sessions and campus-wide hackathons, training and mentoring over 100 students on the ICP Crewsphere platform to foster decentralized application development.